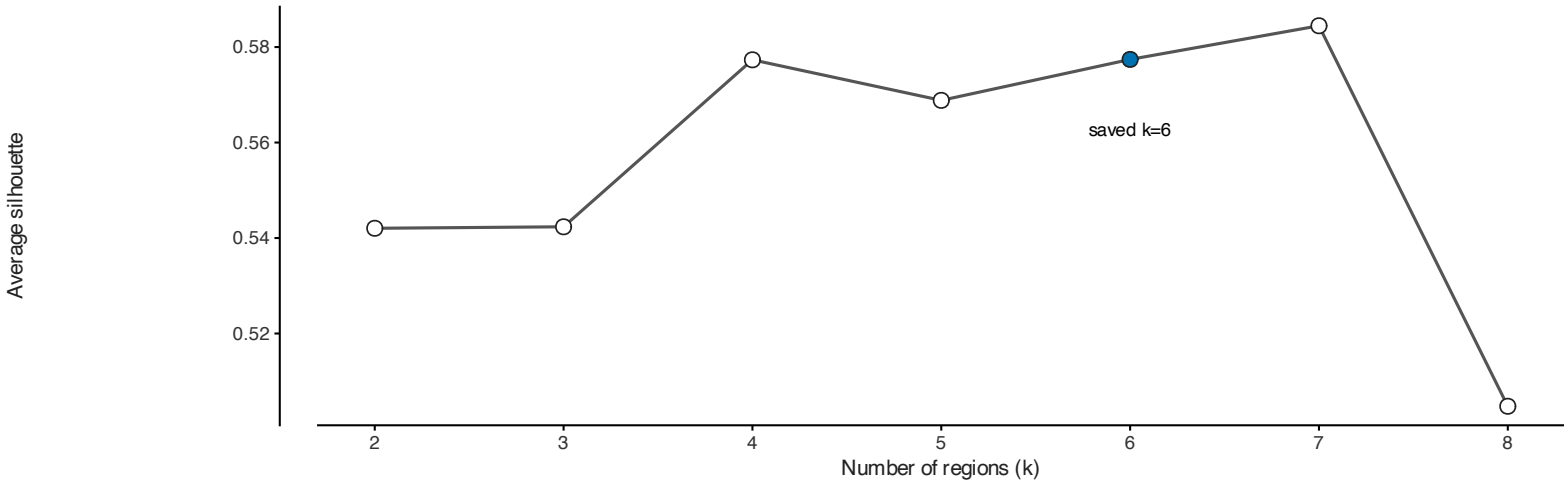


A

Shared primary warm-start-region selection

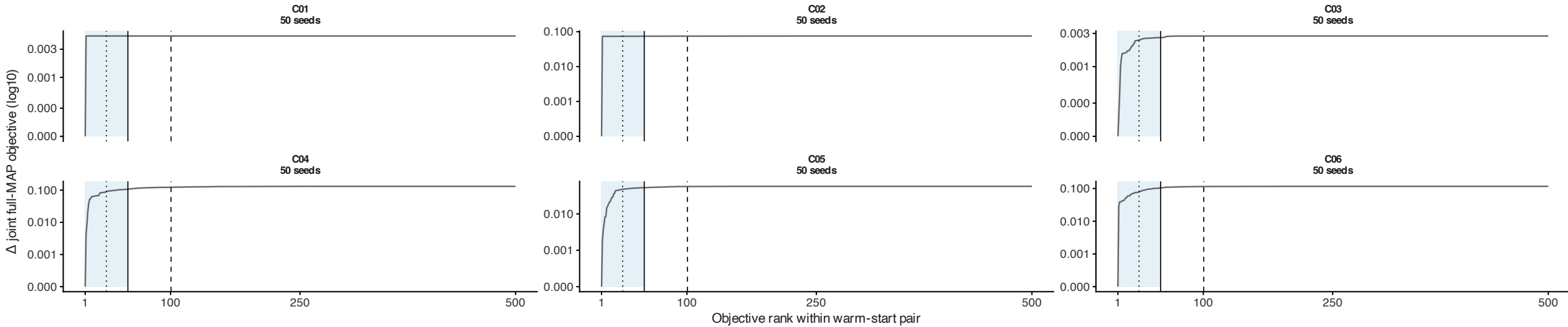
The joint workflow uses the saved six-region primary partition.
80% subsample silhouette maxima: k=2: 6/100; k=4: 27/100; k=5: 2/100; k=6: 33/100; k=7: 26/100; k=8: 6/100.



B

Selected-family joint-objective eligibility

Blue region: lowest 10%; vertical lines mark 5%, 10%, and 20% sets.

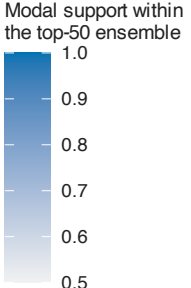


C

Context-specific top-50 endpoint robustness

Text gives the modal result and exact support; a red cross marks a change after collapsing duplicate parameter endpoints.

in vivo	Both ploidy regimes present at 0% O2	yes 100%	no 100%	yes 100%	yes 100%	yes 100%	yes 100%
	Both ploidy regimes present at 1% O2	yes 100%	yes 100%	no 100%	no 100%	yes 100%	yes 100%
	Both ploidy regimes present at 5% O2	no 100%	no 100%	no 100%	no 100%	no 100%	no 100%
	Unmodified trajectory: O2 response direction	decrease 100%	decrease 100%	decrease 100%	decrease 100%	increase 100%	approximately flat 100%
	Surface O2 response at low effective missegregation	increase 100%	increase 100%	increase 100%	increase 100%	increase 100%	increase 100%
	Surface O2 response at intermediate effective missegregation	increase 100%	increase 100%	increase 100%	increase 100%	increase 100%	increase 100%
	Surface O2 response at high effective missegregation	increase 100%	increase 100%	increase 100%	increase 100%	increase 100%	increase 100%
in vitro	At least half of the prespecified diagnostic union has spectral gap ≥ 0.005	yes 100%	no 100%	no 100%	no 100%	yes 100%	yes 100%
	Both ploidy regimes present at 0% O2	yes 100%	yes 100%	yes 100%	yes 100%	yes 100%	yes 100%
	Both ploidy regimes present at 1% O2	yes 100%	yes 100%	yes 100%	yes 100%	yes 100%	yes 100%
	Both ploidy regimes present at 5% O2	yes 100%	yes 100%	yes 100%	yes 100%	yes 100%	yes 100%
	Unmodified trajectory: O2 response direction	approximately flat 100%	approximately flat 100%	approximately flat 100%	approximately flat 100%	approximately flat 100%	approximately flat 100%
	Surface O2 response at low effective missegregation	approximately flat 100%	approximately flat 100%	approximately flat 100%	approximately flat 100%	approximately flat 100%	approximately flat 100%
	Surface O2 response at intermediate effective missegregation	increase 100%	increase 100%	increase 100%	increase 100%	increase 100%	increase 100%
	Surface O2 response at high effective missegregation	decrease 100%	decrease 100%	decrease 100%	decrease 100%	decrease 100%	decrease 100%
	At least half of the prespecified diagnostic union has spectral gap ≥ 0.005	yes 100%	yes 100%	yes 100%	yes 100%	yes 100%	yes 100%
		C01	C02	C03	C04	C05	C06
		Selected warm-start family					



A records the six primary warm-start regions. B-C include C01-C06, the six pairs displayed in Figure 6; optimizer seeds are not biological replicates.